

Geometry of Safety and Utility in Activation Spaces of Aligned Large Language Models

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Abstract

Aligned large language models must refuse harmful instructions while remaining helpful on benign tasks. Yet behavioral guardrails can fail under adversarial prompts without obvious capability collapse, which motivates asking whether safety and helpfulness rely on shared or separable structure inside the network. We study this question under the Linear Representation Hypothesis. From residual-stream activations and contrastive prompts, we estimate safety-aligned and utility-aligned geometry using difference-of-means directions and activation-aware SVD subspaces, pairing refusal-vector-style summaries with richer low-rank subspace analyses emphasized in prior safety work. We apply the same joint safety-utility comparison across several instruction-tuned checkpoints spanning multiple model families, including Gemma, Llama, Qwen, and Phi variants. Anchors are AdvBench (harmful), Alpaca (benign breadth), and HelpSteer (human helpfulness contrasts). Layer by layer we report cosine similarity between DoM directions, plus subspace overlap and principal-angle summaries from actSVD. Selected setups additionally combine these geometric statistics with activation ablations and refusal-focused evaluation so internal overlap can be read alongside behavioral change. We summarize cross-model patterns, connect them to localized refusal and safety-subspace evidence in the literature, and discuss limitations and next steps toward causal interventions.

1 Introduction

Large language models (LLMs) must refuse harmful requests yet remain helpful on benign tasks. Behavioral alignment is fragile: adversarial prompts can bypass safety filters without collapsing general competence (Zou et al., 2023). That contrast suggests a representational question: do safety and ordinary task performance depend on the same internal features, or can they be separated enough for tar-

geted steering? If both behaviors share a bottleneck, improving refusal may disturb instruction following, and the reverse may also hold; if they occupy largely non-overlapping structure, projection-style interventions become more plausible than whole-sale retraining.

We approach the question through mechanistic interpretability and geometry. Under the Linear Representation Hypothesis (LRH), concepts often correspond to directions or low-dimensional subspaces in residual-stream activations (Park et al., 2023). We operationalize *safety* and *utility* as vectors or subspaces estimated from contrastive prompts and ask whether they are *disentangled*, in the sense of low directional overlap or small subspace overlap at a fixed depth, rather than entangled. Throughout, disentanglement denotes a linear geometric claim under fixed data conditions; establishing full causal separability still requires interventions beyond these statistics.

Recent work already localizes safety in activation space. Difference-of-means (DoM) directions capture much refusal variation (Arditi et al., 2024); activation-aware SVD (actSVD) surfaces sparse low-rank safety structure (Wei et al., 2024); and some layers disproportionately separate benign from malicious inputs (Li et al., 2025). How these summaries relate to representations of everyday helpfulness across depth remains comparatively open. We therefore combine one-dimensional DoM summaries with richer actSVD subspaces, reflecting different granularities emphasized in prior safety work, and study safety and utility together.

We extract hidden states from instruction-tuned checkpoints spanning Gemma, Llama, Qwen, and Phi model families. Exact checkpoints appear in Methods. Contrasts use AdvBench (harmful prompts), Alpaca (broad benign instructions), and HelpSteer (human helpfulness ratings) (Wang et al., 2024). Layer by layer we report cosine similarity between DoM directions and overlap and principal-

angle measures between actSVD subspaces. Hook-based ablations and refusal-focused benchmarks complement geometry on selected setups so overlap statistics can be read next to behavioral change rather than in isolation.

Contributions.

- We formulate safety-utility disentanglement in activation space and relate it to refusal-direction, actSVD, and safety-layer analyses (Arditi et al., 2024; Wei et al., 2024; Li et al., 2025).
- We provide a reproducible layer-wise pipeline for DoM directions and actSVD subspaces across multiple aligned models and standard contrastive datasets.
- We combine cross-model layer-wise geometry with ablation-backed behavioral checks where available and sketch extensions toward fuller causal interventions.

The remainder of the report reviews related work, then presents methods, experiments, results, limitations, and discussion.

2 Related Work

Our research builds upon recent advancements in the mechanistic interpretability of Large Language Models (LLMs), specifically focusing on how safety and utility are represented internally.

Vulnerability of Safety Alignment. Current methods for aligning LLMs to be safe rely heavily on behavioral fine-tuning (Zou et al., 2023). However, research has demonstrated that these surface-level guardrails are fundamentally brittle. By using adversarial prompt suffixes, safety filters can be bypassed entirely. This vulnerability strongly motivates our project; because behavioral filters can be easily tricked, we hypothesize that true safety alignment must be engineered at the mechanistic level within the model’s internal representations.

The Linear Representation Hypothesis. The theoretical foundation of our approach is the Linear Representation Hypothesis (LRH) (Park et al., 2023). LRH suggests that high-level concepts occupy specific linear subspaces within a model’s activation space. Previous work demonstrated that making causal interventions on these concept vectors predictably alters the generated text. Our project heavily relies on this methodology. We apply this

geometric framework to test whether intervening on an isolated safety vector leaves the utility capabilities of the model untouched.

Refusal as a Single Direction. Investigating how safety mechanisms operate geometrically, recent studies discovered that an LLM’s refusal behavior against malicious prompts is highly localized. Rather than being scattered across the network, refusal is mediated by a single, one-dimensional subspace (Arditi et al., 2024). This specific "refusal direction" can be successfully isolated using a Difference of Means (DoM) approach. We directly replicate this DoM methodology as our primary baseline to extract safety vectors and test their relationship to utility.

Subspaces and actSVD. Expanding on the geometry of safety, researchers have also mapped the broader representational footprint of these mechanisms (Wei et al., 2024). Safety mechanisms actually occupy a remarkably sparse, low-rank subspace within the model. By applying Activation-aware Singular Value Decomposition (actSVD), it is possible to successfully separate safety-critical regions from utility-critical ones. We adopt this actSVD methodology as our second extraction technique. This allows us to compare the sparse subspace identified by actSVD against the single vector extracted via DoM, helping us robustly map the boundaries between safety and utility representations.

Safety Layers in Aligned LLMs. Finally, in addition to identifying specific directions and subspaces, recent investigations have localized safety mechanisms to specific "safety layers" within aligned LLMs (Li et al., 2025). These layers act as primary filters for distinguishing between benign and malicious prompts. By targeting these specific layers for analysis and ablation, we can better understand where the safety-utility conflict is most intense during the model’s forward pass.

3 Methods

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4 Experiments

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5 Results

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6 Limitations

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7 Discussion and Conclusion

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8 Contribution Statement

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A Appendix

A.1 Sample Size Lower Bound

A possible improvement to extract the utility direction is to see the difference of usefulness=4 and the usefulness mean. This (Obviously) requires the mean of the activations on varied helpfulness scores on HelpSteer (Wang et al., 2024). To do this we first take a sample of HelpSteer to estimate the mean,

in this section we aim to find a lower bound for the sample size to get a good estimate of the mean.

Lemma 1. *Let $G = \{g_1, \dots, g_5\}$ be the set of groups, and choose a group $g_k \in G$ that has the distribution P_{g_k} . Then let $x_1, \dots, x_n \stackrel{\text{iid}}{\sim} P_{g_k}$ be samples from g_k , and let $a_i = f(x_i)$ be the activations produced by passing each x_i through the network. Then $a_1, \dots, a_n$ are i.i.d*

Proof. The a_i are identically distributed because each x_i is drawn from the same distribution P_{g_k} and each a_i is produced by passing forward i.i.d samples on the exact same model, therefore each a_i is drawn from the same distribution of activations. Further, computing $f(x_i)$ shares no state whatsoever with computing $f(x_j)$ for $i \neq j$ and thus a_i and a_j are independent $\square$

Note: For the previous lemma to hold we require that each activation is sampled the same way, we extract activations from the last token so this was omitted as it is not necessary to list that condition for our case.

Theorem 1. *Let $\epsilon, \delta > 0$ then we sample $n_k = p_k N$ i.i.d samples from each group $g_k \in G$, and p_k is the proportion of g_k in HelpSteer. We pass the samples and construct an estimator of the mean $\hat{\mu} = \sum_{k=1}^5 p_k \bar{a}_k$. Then if $N \geq \frac{\text{tr}(\Sigma)}{\delta \epsilon^2}$ we have that $P(\|\mu - \hat{\mu}\| < \epsilon) \geq 1 - \delta$ where $\Sigma = \sum_k p_k \Sigma_k$*

Proof. By lemma we have that the activations from each group $g_k \in G$ are i.i.d, so we have that $\mathbb{E}[\bar{a}_k] = \mu_k$ and then using the linearity of expectation we get that

$\mathbb{E}[\sum_k p_k \bar{a}_k] = \sum_k p_k \mathbb{E}[\bar{a}_k] = \sum_k p_k \mu_k = \mu$ so the mean we have is not biased. Next we want to find the variance of $\hat{\mu}$, we find this by using our lemma, we have that $\text{Var}(\bar{a}_k) = \frac{\Sigma_k}{n_k}$, further, $\text{Var}(\hat{\mu}) = \sum_k \text{Var}(p_k \bar{a}_k) = \sum_k p_k^2 \frac{\Sigma_k}{n_k}$ using the definition of $\hat{\mu}$, $\bar{a}$ and basic property of the variance. Then we substitute $n_k = p_k N$ and get

$\sum_k p_k \frac{\Sigma_k}{N} = \frac{\bar{\Sigma}}{N}$ by moving the constant out and taking the sum. So, $\text{Var}(\hat{\mu}) = \frac{\bar{\Sigma}}{N}$, finally we apply Chebyshev’s inequality and put $\text{Var}(\hat{\mu}) = \frac{\bar{\Sigma}}{N}$ and get that

$P(\|\hat{\mu} - \mu\| < \epsilon) \geq 1 - \frac{\bar{\Sigma}}{N \epsilon^2}$, and then solving for N gives us $N \geq \frac{\text{tr}(\bar{\Sigma})}{\delta \epsilon^2}$ $\square$